L^2-BETTI NUMBERS FROM ALGEBRAIC POINT OF VIEW

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Abstract. Fifty years ago, Atiyah asked a question that is now known as the following Atiyah Conjecture: for any group G, are the L^2-Betti numbers of any G-CW-complex always rational?

This short course will focus on Linnell's proof of the (strong) Atiyah Conjecture for a large and concrete class of groups. His proof is based on an elegant algebraization of L^2-Betti numbers via Lück's dimension theory.

For a torsion-free group G, the Atiyah Conjecture predicts the existence of a canonical division ring $\mathcal{D}G$ extending the group ring $\mathbb{Z}G$. The torsion theory over $\mathcal{D}G$ yields the universal L^2-torsion. If time permits, we will demonstrate how the universal L^2-torsion detects the fiberedness of most 3-manifolds.

1. Lecture 1

One thing that a topologist, an algebraist and a group theorist would have in common is they love to study a group G by looking at its action on a mathematical object. Especially:

- $G \curvearrowright A$ where A is an abelian group. This makes A a $\mathbb{Z}G$*-module*.
- $G \curvearrowright X$ cellularly where X is a CW-complex. This makes X a *G-CW complex*. In particular, the homology group of X is a $\mathbb{Z}G$-module.

When G is infinite, the topology of X and its homology group group $H_*(X)$ is typically not of finite type. An idea is to construct the " $\mathbb{Z}G$-dimension" of such modules.

Question 1.1. *Define a "$\mathbb{Z}G$-dimension" for every $\mathbb{Z}G$-module M, such that:*

(1) $\dim_G(\mathbb{Z}G)^n = n$.

(2) Suppose $0 \to M_1 \to M \to M_2 \to 0$ *is exact, then*

$$\dim_G M = \dim_G M_1 + \dim_G M_2.$$

The desired $\mathbb{Z}G$-dimension does not exist for a simple reason:

Example 1.2. Consider the free group of rank two $F = \langle g_1, g_2 \rangle$. Identify F with the fundamental group of $S^1 \vee S^1$, then F naturally acts on the universal cover X, a infinite tree of degree 4.

The cellular chain complex of X is finitely generated as $\mathbb{Z}F$-module.

$$C_*(X) = 0 \to \mathbb{Z}F \oplus \mathbb{Z}F \xrightarrow{\begin{pmatrix} g_1 - 1 \\ g_2 - 1 \end{pmatrix}} \mathbb{Z}F \to 0.$$

It is easy to see that $\partial : \mathbb{Z}F \oplus \mathbb{Z}F \to \mathbb{Z}F$ is injective. In fact, any chain of $C_1(X)$ is supported on finitely many edges, hence have non-trivial endpoints. Similarly,

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there is an embedding of $\mathbb{Z}F$-modules $\mathbb{Z}F \oplus \mathbb{Z}F \hookrightarrow \mathbb{Z}F$ for any free group F of rank greater than 1.

In fact, we have the following theorem. The if-part is proved by Ceccherini-Silberstein and Coornaert [CSC06], the only-if-part is proved by Bartholdi [Bar19] and also Kropholler–Lorensen [KL19].

Theorem 1.3. *A group G is non-amenable if and only if there exists $n > 0$ such that $\mathbb{Z}G^{n+1} \hookrightarrow \mathbb{Z}G^n$.*

This suggests that the ring $\mathbb{Z}G$ is algebraically highly complicated. However, we do have a beautiful dimension theory if we pass to the Hilbert space completion of $\mathbb{Z}G$.

1.1. **von Neumann dimension.**

Definition 1.4. Let G be a group. Define $\ell^2(G)$ to be the Hilbert space spanned by elements of G. Namely,

$$\ell^2(G) = \Big\{ \sum_{g\in G} c_g \cdot g \;\Big|\; c_g \in \mathbb{C},\ \sum_{g\in G} |c_g|^2 < \infty \Big\}.$$

The inner product is given by

$$\Big\langle \sum_{g\in G} c_g \cdot g, \sum_{g\in G} d_g \cdot g \Big\rangle = \sum_{g\in G} c_g \overline{d_g}.$$

There is a natural isometric G-action on both sides of $\ell^2(G)$.

We wish to define the dimension of the following class of spaces:

Definition 1.5 (Hilbert G-module)**.** A finitely generated *Hilbert G-module* is a Hilbert space V together with an isometric left G-linear action, such that there exists an isometric G-embedding

$$V \hookrightarrow \ell^2(G)^n$$

for some n. The *morphisms* between Hilbert G-modules are bounded G-linear maps.

Example 1.6. Any closed G-invariant subspace of $\ell^2(G)^n$ is a f.g. Hilbert G-module. Conversely, any f.g. Hilbert G-module is G-linearly isomorphic to a closed G-invariant subspace of $\ell^2(G)^n$. Therefore

$$\{\text{Hilbert } G\text{-modules}\}/\sim \quad \overset{1:1}{\longleftrightarrow} \quad \{\text{closed } G\text{-invariant subspaces of } \ell^2(G)^n\}/\sim .$$

Definition 1.7 (von Neumann algebra)**.** Define the von Neumann algebra

$$\mathcal{N}G = \{\text{bounded } G\text{-linear operators } f : \ell^2(G) \to \ell^2(G)\}.$$

Remark 1.8. It is easy to verify the following facts:

(1) $\mathcal{N}G$ is a $\mathbb{C}$-algebra. In particular $\mathbb{Z}G \subset \mathcal{N}G$ by right multiplication.
(2) An operator $f \in \mathcal{N}G$ is determined by the value $f(e) \in \ell^2(G)$. Therefore, as a set,

$$\mathcal{N}G = \{x \in \ell^2(G) \mid ax \in \ell^2(G),\ \forall a \in \ell^2(G)\}.$$

(3) Note that $\ell^2(G)$ is in general not a ring.

Theorem 1.9 (L^p-conjecture, [Sae90])**.** *Suppose G is a infinite discrete group, then $\ell^p(G) \neq \ell^p(G) \cdot \ell^p(G)$ for any $p > 1$.*

Suppose we enumerate the element of G as $\{g_i\}_{i \in I}$, then the matrix $\mathcal{A}$ representing $f \in \mathcal{N}G$ has the form

$$\mathcal{A}_{ij} = \langle f(g_i), g_j \rangle = \langle g_i f(e), g_j \rangle = \langle f(e), g_i^{-1} g_j \rangle.$$

Hence the diagonal elements are constantly $\mathrm{tr}_{\mathcal{N}G}(f)$.

Definition 1.10 (von Neumann trace)**.** The von Neumann trace is a $\mathbb{C}$-linear map $\mathrm{tr}_{\mathcal{N}G} : \mathcal{N}G \to \mathbb{C}$ defined as

$$\mathrm{tr}_{\mathcal{N}G}\, f := \langle f(e), e \rangle, \quad \forall f \in \mathcal{N}G.$$

There are at least two natural topologies on $\mathcal{N}G$, all inherited from $\mathcal{B}(\ell^2(G))$: the norm topology, and the weak topology. The von Neumann trace is continuous with respect to both topology. Moreover, as a trace function, we certainly have

$$\mathrm{tr}_{\mathcal{N}G}(fg) = \mathrm{tr}_{\mathcal{N}G}(gf), \quad f, g \in \mathcal{N}G.$$

The trace can be used to characterize the dimension of a vector space. Similarly, we have the following definitions.

Definition 1.11 (von Neumann dimension)**.** Suppose $V \subset \ell^2(G)$ is a closed G-invariant subspace. Let $\mathrm{pr} : \ell^2(G) \to \ell^2(G)$ be the orthogonal projection onto V. Define the *von Neumann dimension* of V as

$$\dim_{\mathcal{N}G} V := \mathrm{tr}_{\mathcal{N}G}(\mathrm{pr}) \in [0, 1].$$

Similarly, we define the von Neumann dimension for any closed G-invariant subspace $V \subset \ell^2(G)^n$:

$$V \longmapsto \dim_{\mathcal{N}G} V \in [0, n].$$

Finally, define the von Neumann dimension of a fin. gen. Hilbert G-module M using the embedding $M \hookrightarrow \ell^2(G)^n$.

Remark 1.12. (1) The von Neumann dimension is independent of the choice of the embedding.

Proof. Assume that there are two isometric G-embeddings i, j and projections A, B, respectively.

$$\begin{tikzcd} \ell^2(G)^n \arrow[r, bend left, "A"] \arrow[r, shift left, "j"] & M \arrow[l, shift left, "i"] \arrow[l, shift right] \arrow[l, bend right, "B"] \end{tikzcd}$$

Then the isomorphism $\mathrm{im}\, A \cong \mathrm{im}\, B$ trivially extends to a morphism $U : \ell^2(G)^n \to \ell^2(G)^n$ by zero on the orthogonal complements. Therefore $j = U \circ i$, $A = B \circ U$, and.

$$\mathrm{tr}_{\mathcal{N}G}(j \circ B) = \mathrm{tr}_{\mathcal{N}G}(U \circ i \circ B) = \mathrm{tr}_{\mathcal{N}G}(i \circ B \circ U) = \mathrm{tr}_{\mathcal{N}G}(i \circ A).$$

□

(2) (Regularity) For any fin. gen. Hilbert G-module M,

$$\dim_{\mathcal{N}G} M = 0 \iff M = 0.$$

(3) (Additivity) If $0 \to M_1 \to M \to M_2 \to 0$ is weakly exact, then

$$\dim_{\mathcal{N}G} M = \dim_{\mathcal{N}G} M_1 + \dim_{\mathcal{N}G} M_2.$$

Example 1.13. Consider the free abelian group $\mathbb{Z}^n$. By standard Fourier transformation, we can identify $\ell^2(\mathbb{Z}^n)$ with $L^2(T^n)$, the complex Hilbert space of L^2-integrable functions on the n-dimensional torus $T^n = (S^1)^n$. The basis element $(k_1, \dots, k_n) \in \mathbb{Z}^n$ is sent to the function

$$f_{k_1,\dots,k_n} : T^n \to \mathbb{C}, \quad (z_1, \dots, z_n) \mapsto z_1^{k_1} \cdots z_n^{k_n}.$$

The inner product of $f, g \in L^2(T^n)$ is given by

$$\langle f, g \rangle := \frac{1}{(2\pi)^n} \int_{T^n} f(w)\overline{g(w)} dw.$$

The von Neumann algebra $\mathcal{N}\mathbb{Z}^n$ has the description

$$\begin{aligned} \mathcal{N}\mathbb{Z}^n &= \{x \in \ell^2(\mathbb{Z}^n) \mid ax \in \ell^2(\mathbb{Z}^n),\ \forall a \in \ell^2(\mathbb{Z}^n)\} \\ &= \{f \in L^2(T^n) \mid gf \in L^2(T^n),\ \forall g \in L^2(T^n)\} \\ &= L^\infty(T^n). \end{aligned}$$

The von Neumann trace $\mathrm{tr}_{\mathcal{N}\mathbb{Z}^n} : \mathcal{N}\mathbb{Z}^n \to \mathbb{C}$ is given by

$$\mathrm{tr}_{\mathcal{N}\mathbb{Z}^n} : L^\infty(T^n) \to \mathbb{C}, \quad \mathrm{tr}_{\mathcal{N}\mathbb{Z}^n}(f) := \langle f, 1 \rangle = \frac{1}{(2\pi)^n} \int_{T^n} f.$$

A closed G-invariant subspace $M \subset \ell^2(\mathbb{Z}^n)$ is characterized by an orthogonal projection $\mathrm{pr} \in \mathcal{N}\mathbb{Z}^n$. Namely, pr is an self-adjoint idempotent. These corresponds to the characteristic functions

$$\chi_A : T^2 \to \mathbb{C}, \quad A \subset T \text{ measurable set.}$$

The von Neumann dimension of M is given by $\mathrm{Vol}(A) \in [0, 1]$. In conclusion,

$\ell^2(\mathbb{Z}^n)$	$\mathcal{N}\mathbb{Z}^n$	$\mathrm{tr}_{\mathcal{N}\mathbb{Z}^n}$	submodule $M \subset \ell^2(\mathbb{Z}^n)$	$\dim_{\mathcal{N}\mathbb{Z}^n} M$
$L^2(T^n)$	$L^\infty(T^n)$	$f \mapsto \int_{T^n} f$	measurable set $A \subset T^n$	$\mathrm{Vol}(A)/(2\pi)^n$

Example 1.14. Recall from Example 1.2 the F-CW complex X for free group $F = \langle g_1, g_2 \rangle$.

$$C_*(X) = 0 \to \mathbb{Z}F \oplus \mathbb{Z}F \xrightarrow{\begin{pmatrix} g_1 - 1 \\ g_2 - 1 \end{pmatrix}} \mathbb{Z}F \to 0.$$

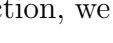
If we pass to the L^2-completion, i.e. we allow the cellular chains to be infinite:

$$C_*^{(2)}(X) = 0 \to \ell^2(F) \oplus \ell^2(F) \xrightarrow{\begin{pmatrix} g_1 - 1 \\ g_2 - 1 \end{pmatrix}} \ell^2(F) \to 0.$$

Observe that:

- $\operatorname{im}\partial$ contains the augmentation ideal $\mathcal{I} \subset \mathbb{Z}F \subset \ell^2(F)$, a dense subset of $\ell^2(F)$. Therefore
$$\overline{\operatorname{im}\partial} = \ell^2(F).$$
(In fact, the 0-th L^2-Betti number of any CW-complex with infinite fundamental group is zero.)
- $0 \to \ker\partial \to \ell^2(F) \oplus \ell^2(F) \xrightarrow{\partial} \overline{\operatorname{im}\partial} \to 0$ is weakly exact, therefore
$$\dim_{\mathcal{N}G}(\ker\partial) = 1.$$

Definition 1.15 (L^2-homology)**.** Let G acts freely, cellularly and cocompactly on a CW-complex X. Define
$$\begin{aligned} C_*^{(2)}(X) &= \ell^2(G) \otimes_{\mathbb{Z}G} C_*(X), \quad (L^2\text{-cellular chain complex}) \\ H_i^{(2)}(X) &= \ker\partial_i^{(2)}/\overline{\operatorname{im}\partial_{i+1}^{(2)}}, \quad (L^2\text{-homology, a Hilbert } G\text{-module}) \\ \beta_i^{(2)}(X) &= \dim_{\mathcal{N}G} H_i^{(2)}(X) \quad (L^2\text{-Betti numbers}). \end{aligned}$$
In the special case that X is the universal cover of a finite connected CW-complex Y with obvious $\pi_1(Y)$-action, we often abbreviate the above by the corresponding invariants of Y.

Example 1.16. When $Y =$  , we have $\beta_1^{(2)}(Y) = 1$, $\beta_0^{(2)}(Y) = 0$.

There are many standard topological properties of the L^2-homology: *homotopy invariance*, *Künneth formula*, *Poincaré duality*, *Euler–Hopf formula*, and so on. These can be found in any textbook on L^2-invariants. Actually one can even guess their meaning from the name.

1.2. **Extended von Neumann dimension.** There are two restrictions in the above definition of L^2-Betti numbers.

(1) We would like a more algebraic-flavored definition of L^2-Betti number, instead of caring about functional analysis.
(2) There are lots of interesting CW-complex with group action which is not cocompact, such as $G \curvearrowright EG$ where EG is the universal cover of the classifying space BG. Therefore, we have to study Hilbert G-modules which is *not* finitely generated.

In the above discussion, we were working in the following category:

Definition 1.17. The category $\mathcal{C}_1$ of finitely generated Hilbert G-module is defined as follows:
$$\begin{aligned} \operatorname{Obj}(\mathcal{C}_1) &= \{\text{fin. gen. Hilbert } G\text{-modules}\}, \\ \operatorname{Mor}(\mathcal{C}_1) &= \{\text{bounded } G\text{-linear maps}\}. \end{aligned}$$

View $\mathcal{N}G$ as a abstract ring, we have the following category:

Definition 1.18. The category $\mathcal{C}_2$ of finitely generated projective $\mathcal{N}G$-module is defined as follows:
$$\begin{aligned} \operatorname{Obj}(\mathcal{C}_2) &= \{\text{fin. gen. projective } \mathcal{N}G\text{-modules}\}, \\ \operatorname{Mor}(\mathcal{C}_2) &= \{\mathcal{N}G\text{-linear maps}\}. \end{aligned}$$

Theorem 1.19. *The categories $\mathcal{C}_1, \mathcal{C}_2$ are equivalent. Namely, there are functors*

$$\nu : \mathcal{C}_2 \to \mathcal{C}_1 \quad \text{and} \quad \nu^{-1} : \mathcal{C}_1 \to \mathcal{C}_2$$

inverses to each other up to natural equivalences.

Proof. The proof roughly goes as follows: construct the functor ν_1 that sends the regular Hilbert G-modules $\ell^2(G)^n$ to the free $\mathcal{N}G$-module $\mathcal{N}G^n$. A morphism $A : \ell^2(G)^n \to \ell^2(G)^m$ is a matrix over $\mathcal{N}G$, hence naturally a morphism $\nu^{-1}(A) : \mathcal{N}G^n \to \mathcal{N}G^m$. And vise versa.

Generally, a finitely generated Hilbert G-module can be identified with a pair $(\ell^2(G)^n, p)$, where $p : \ell^2(G)^n \to \ell^2(G)$ is an *idempotent*, namely p is a idempotent matrix over $\mathcal{N}G$. The functor ν sends the pair $(\ell^2(G)^n, p)$ to $(\mathcal{N}G^n, p)$, identifying a projective $\mathcal{N}G$-module $p(\mathcal{N}G^n)$. □

Definition 1.20 (Extended von Neumann dimension)**.** Define the *extended von Neumann dimension* $\dim_{\mathcal{N}G}(M)$ for arbitrary $\mathcal{N}G$-module M:

- If M is finitely generated projective, define
$$\dim_{\mathcal{N}G} M := \dim_{\mathcal{N}G}(\nu(M)).$$
- If M is any $\mathcal{N}G$-module. Define its *extended von Neumann dimension*
$$\dim_{\mathcal{N}G}(M) := \sup\{\dim_{\mathcal{N}G} P \mid P \subset M \text{ fin. gen. projective submodule}\} \in [0, +\infty].$$

Definition 1.21 (Extended L^2-Betti number)**.** Let X be a topological space, with continuous G-action. The homology of X with coefficient in $\mathcal{N}G$

$$H_*(X; \mathcal{N}G)$$

is the homology of the $\mathcal{N}G$-chain complex

$$\mathcal{N}G \otimes_{\mathbb{Z}G} C_*^{\text{sing}}(X).$$

The L^2-Betti number of X is given by

$$\beta_p^{(2)}(X) := \dim_{\mathcal{N}G}(H_p^G(X; \mathcal{N}G)) \in [0, \infty].$$

The L^2-Betti number of a discrete group G is defined by

$$\beta_p^{(2)}(G) := \beta_p^{(2)}(EG; \mathcal{N}G).$$

Remark 1.22. (1) When X is a free, cocompact G-CW complex, then the extended L^2-Betti number agrees with the original one.

(2) Given a chain complex

$$\cdots \to \ell^2(G)^{r_{n+1}} \xrightarrow{\partial_{n+1}} \ell^2(G)^{r_n} \xrightarrow{\partial_n} \ell^2(G)^{r_{n-1}} \to \cdots.$$

The usual homology is not a Hilbert space. We have to define

$$H_n^{(2)} := \ker \partial_n / \overline{\operatorname{im} \partial_{n+1}}.$$

However, when viewed as chain complex of von Neumann algebras

$$\cdots \to \mathcal{N}G^{r_{n+1}} \xrightarrow{\partial_{n+1}} \mathcal{N}G^{r_n} \xrightarrow{\partial_n} \mathcal{N}G^{r_{n-1}} \to \cdots.$$

The classical homology is a well-defined $\mathcal{N}G$-module (though not projective in general)

$$H_n := \ker \partial_n / \operatorname{im} \partial_{n+1}.$$

And we have $\dim_{\mathcal{N}G} H_n = \dim_{\mathcal{N}G} H_n^{(2)}$.

(3) The tradeoff is that for an arbitrary $\mathcal{N}G$-module M,

$$\dim_{\mathcal{N}G} M = 0 \quad \text{does not imply} \quad M = 0.$$

Lück [Lüc02, Section 6] had a belief that the von Neumann algebra is very similar to the ring $\mathbb{Z}$ of integers with two exceptions: (when G is infinite) $\mathcal{N}G$ is not Noetherian, and $\mathcal{N}G$ has non-trivial zero-divisors. The von Neumann dimension of a $\mathcal{N}G$-module is comparable with the $\mathbb{Z}$-rank of a $\mathbb{Z}$-module. In the next lecture we will consider the "localization" of $\mathcal{N}G$: the affiliated algebra $\mathcal{U}G$. Naturally, $\mathcal{U}G$ to $\mathcal{N}G$ is like $\mathbb{Q}$ to $\mathbb{Z}$.

Let me mention the following beautiful theorem.

Theorem 1.23 (Lück [Lüc98], Kropholler–Lorensen [KL19])**.** *A group G is amenable if and only if*

$$\dim_{\mathcal{N}G}(\mathrm{Tor}_p^{\mathbb{C}G}(M, \mathcal{N}G)) = 0$$

for all $p \geqslant 0$ and all $\mathbb{Z}G$-module M.

Corollary 1.24. *If G is an infinite amenable group, then the L^2-Betti numbers of G vanish.*

2. Lecture 2

2.1. **Atiyah Conjecture.** Fix a group G. Let's look at the set of all possible L^2-Betti numbers.

Proposition 2.1. *For any group G. The following sets are the same:*

(1) $\Lambda_1 := \{\beta_p^{(2)}(M; \mathcal{N}G) \mid M$ *is a free, cocompact G-manifold,* $p \in \mathbb{Z}\}$.
(2) $\Lambda_2 := \{\beta_p^{(2)}(X; \mathcal{N}G) \mid X$ *is a free, cocompact G-CW complex,* $p \in \mathbb{Z}\}$.
(3) $\Lambda_3 := \{\dim_{\mathcal{N}G}(\ker r_A) \mid r_A : \ell^2(G)^n \to \ell^2(G)^m$ *given by a matrix A over $\mathbb{Z}G$*$\}$.
(4) $\Lambda_4 := \{\dim_{\mathcal{N}G}(\mathcal{N}G \otimes_{\mathbb{Z}G} M) \mid M$ *is a finitely presented $\mathbb{Z}G$-module*$\}$.

Therefore, we denote this set by $\Lambda(G) \subset [0, +\infty)$.

Proof.
- $\Lambda_1 \subset \Lambda_2$: this is obvious.
- $\Lambda_2 \subset \Lambda_1$: embed $G\backslash X$ into $\mathbb{R}^N$ for N large enough. Let M be the regular neighborhood of $G\backslash X \subset \mathbb{R}^N$. Then ∂M is a closed manifold and the projection from the ∂M to $G\backslash X$ is $(N - \dim X - 2)$-connected.
- $\Lambda_2 = \Lambda_3$: the support of A is a finitely generated subgroup $H < G$. Realize A as a boundary map of an H-CW complex Y, then consider $X = G \times_H Y$.
- $\Lambda_3 = \Lambda_4$: suppose M is finitely presented $\mathbb{Z}G$-module with

$$\mathbb{Z}G^n \xrightarrow{r_A} \mathbb{Z}G^m \to M \to 0$$

exact. Then the $\mathcal{N}G$-complex

$$\mathcal{N}G^n \xrightarrow{r_A} \mathcal{N}G^m \to \mathcal{N}G \otimes_{\mathbb{Z}G} M \to 0$$

is also exact, with

$$\dim_{\mathcal{N}G}(\mathcal{N}G \otimes_{\mathbb{Z}G} M) = m - n + \dim_{\mathcal{N}G}(\ker r_A).$$

□

Remark 2.2. The proposition states for all groups, but essentially reduces to finitely generated groups.

Conjecture 2.3 (Atiyah [Ati76]).

$$\Lambda(G) \subset \mathbb{Q}.$$

Not surprisingly, this conjecture is not true if we put no restriction on the group.

Theorem 2.4 (Grabowski [Gra14], Pichot–Schick–Zuk [PSZ15]).

$$\bigcup_{\text{fin. gen. } G} \Lambda(G) = \mathbb{R}_{\geqslant 0}.$$

Conjecture 2.5 (Strong Atiyah Conjecture). *Let N be the least common multiple of the order of finite subgroups of G. Then*

$$\Lambda(G) \subset \frac{1}{N} \cdot \mathbb{Z}.$$

In particular, if G is torsion-free, then

$$\Lambda(G) \subset \mathbb{Z}.$$

Conjecture 2.6 (Kaplansky zero divisor conjecture). *Suppose G is torsion-free, then $\mathbb{Z}G$ contains no nontrivial zero divisors.*

For torsion-free groups, the Atiyah Conjecture implies the Kaplansky Conjecture as follows: for any $x \in \mathbb{Z}G$, if x is a non-zero divisor, then $r_x : \ell^2(G) \to \ell^2(G)$ has non-trivial kernel, hence

$$\dim_{\mathcal{N}G} \ker r \in \mathbb{Z}_+,$$

therefore $\dim_{\mathcal{N}G} \ker r = 1$. This shows that $r_x = 0$, and $x = r_x(e) = 0$.

2.2. **Linnell's class $\mathcal{C}$.** The main goal of this course is the Atiyah Conjecture for the following class of groups:

Definition 2.7 (Linnell's class $\mathcal{C}$). Let $\mathcal{C}$ be the smallest class of groups which contains all free groups and is closed under:

- directed unions, and
- extensions by elementary amenable groups.

We remark that $\mathcal{C}$ is closed under *taking subgroups*, closed under any *free products*, and contains all *3-manifold groups* [KL24].

Theorem 2.8 (Linnell [Lin93]). *Let G be a group in $\mathcal{C}$ and suppose there is a upper bound on the orders of finite subgroups of G. Then G satisfies the strong Atiyah Conjecture.*

The proof is complicated and the key is to deal with finite extensions within the class $\mathcal{C}$. The following general assertion remains open:

Conjecture 2.9 ([LS07]). *Suppose K satisfies the Strong Atiyah Conjecture and G is a finite extension of K. Then G satisfies the Atiyah Conjecture.*

2.3. **The algebra of affiliated operators.** We will need the following important construction. Intuitively,

$$\text{Hilbert } \mathcal{N}G\text{-module} \rightsquigarrow \mathcal{N}G\text{-module} \rightsquigarrow \mathcal{U}G\text{-module}$$
$$\mathbb{Z}^n \rightsquigarrow \mathbb{Z}\text{-module} \rightsquigarrow \mathbb{Q}\text{-module}$$

Let V, W be a Hilbert space. An *(unbounded)* operator $f : V \to W$ is a linear map defined on its domain $\mathrm{dom}(f) \subset V$. The operator f is called *closed* if its graph $\{(x, f(x)) \in V \times W \mid x \in \mathrm{dom}(f)\}$ is a closed subset.

Theorem 2.10 (Closed graph theorem)**.** *If a closed linear operator $f : \mathrm{dom}(f) \subset U \to V$ is everywhere defined, then f is bounded.*

This explains the name "unbounded operator": if an closed operator is unbounded, then its domain must be a proper subset.

Given $f \in \mathcal{N}G$, recall that x is by definition a bounded G-linear operator

$$f : \ell^2(G) \to \ell^2(G).$$

Suppose f is injective, then f is a weak isomorphism and $\mathrm{im}\, f$ is dense. The inverse $f^{-1} : \ell^2(G) \to \ell^2(G)$ is an unbounded G-linear operator, defined on the dense subset $\mathrm{im}\, f$. Moreover, the graph of f^{-1} is the same as f, hence f^{-1} is closed.

Definition 2.11 (Affiliated operators)**.** Define $\mathcal{U}G$ to be the set of densely defined, closed, (unbounded) G-linear operators.

Proposition 2.12. *The basic properties of $\mathcal{U}G$:*

(1) *Addition and composition of affiliated operators are well-defined. For any $f, g \in \mathcal{U}G$, define $(f + g) \in \mathcal{U}G$ as the closure of*
$$x \longmapsto f(x) + g(x), \quad x \in \mathrm{dom}(f) \cap \mathrm{dom}(g),$$
define $f \circ g \in \mathcal{U}G$ as the closure of
$$x \longmapsto f(g(x)), \quad x \in g^{-1}(\mathrm{dom}(f)).$$
Verifying the well-definedness is not obvious, but standard. Therefore, $\mathcal{U}G$ becomes a ring.

(2) *Any $f \in \mathcal{U}G$ can be written as $f = ab^{-1}$, where $a, b \in \mathcal{N}G$ and b is injective.*

(3) *$\mathcal{U}G$ is von Neumann regular. Equivalently, any finitely presented submodule of $\mathcal{U}G^n$ is a direct summand* [Lüc02, Lemma 8.18].

The properties (2) and (3) appears natural if one compares $\mathcal{N}G \subset \mathcal{U}G$ with $\mathbb{Z} \subset \mathbb{Q}$. Next, we extend the von Neumann dimension from $\mathcal{N}G$-modules to $\mathcal{U}G$-modules.

Definition 2.13 (Extended von Neumann dimensions)**.** For any $\mathcal{U}G$-module M, define the von Neumann dimension

$$\dim_{\mathcal{U}G} M \in [0, +\infty]$$

as follows:

- It is known that any finitely generated projective $\mathcal{U}G$-module M can be expressed as $M = \mathcal{U}G \otimes_{\mathcal{N}G} N$ where N is a finitely generated projective $\mathcal{N}G$-module [Lüc02, Theorem 8.22]. Define $\dim_{\mathcal{U}G} M := \dim_{\mathcal{N}G} N$.

- For general $\mathcal{U}G$-module M, define
$$\dim_{\mathcal{U}G} M := \limsup\{\dim_{\mathcal{U}G} P \mid P \subset M \text{ fin. gen. projective}\}.$$

This extended von Neumann dimension has all nice properties like the one defined over $\mathcal{N}G$-modules. Importantly [Lüc02, Theorem 8.29], for any $\mathcal{N}G$-module M,
$$\dim_{\mathcal{U}G}(\mathcal{U}G \otimes_{\mathcal{N}G} M) = \dim_{\mathcal{N}G} M.$$

2.4. **Linnell's theorem.** In this note, a field always refers to a *skew field*, a ring whose non-zero elements are all units. Other authors may call it a *division ring* or *division algebra*.

We are ready to prove the following beautiful criterion of Linnell:

Theorem 2.14 (Linnell [Lin93])**.** *Let G be a torsion-free group. Then the Atiyah Conjecture holds for G if and only if there exists a field F with $\mathbb{Z}G \subset F \subset \mathcal{U}G$.*

Proof. Suppose there is a field F with $\mathbb{Z}G \subset F \subset \mathcal{U}G$. Given any finitely presented $\mathbb{Z}G$-module M, we have
$$\begin{aligned}\dim_{\mathcal{N}G}(\mathcal{N}G \otimes_{\mathbb{Z}G} M) &= \dim_{\mathcal{U}G}(\mathcal{U}G \otimes_{\mathbb{Z}G} M)\\ &= \dim_{\mathcal{U}G}(\mathcal{U}G \otimes_F F \otimes_{\mathbb{Z}G} M)\\ &= \dim_{\mathcal{U}G}(\mathcal{U}G \otimes_F F^n) \quad \text{for some } n \in \mathbb{N}\\ &= \dim_{\mathcal{U}G}(\mathcal{U}G^n) = n.\end{aligned}$$
Therefore $\Lambda(G) \subset \mathbb{Z}$.

On the other hand, suppose that G satisfies the Atiyah Conjecture, we prove that there is a field between $\mathbb{Z}G \subset \mathcal{U}G$.

Definition 2.15 (Division closure)**.** The *division closure* of $\mathbb{Z}G \subset \mathcal{U}G$ is the smallest ring $\mathcal{D}G$ such that

(1) $\mathbb{Z}G \subset \mathcal{D}G \subset \mathcal{U}G$, and
(2) $\mathcal{D}G$ is *division-closed*, i.e. any $x \in \mathcal{D}G$ that is invertible in $\mathcal{U}G$ is already invertible in $\mathcal{D}G$.

It is easy to see that a field between $\mathbb{Z}G \subset \mathcal{U}G$ exists if and only if $\mathcal{D}G$ is a field.

To prove $\mathcal{D}G$ is a field, suppose the contrary that $\exists x \in \mathcal{D}G \setminus \{0\}$ not invertible in $\mathcal{U}G$. Consider the right multiplication by x
$$r_x : \mathcal{U}G \to \mathcal{U}G.$$
Since we have exact sequence
$$0 \to \ker r_x \to \mathcal{U}G \to \operatorname{im} r_x \to 0,$$
the *finitely generated* submodule $\operatorname{im}(r_x)$ is direct summand of $\mathcal{U}G$ by von Neumann regular. Hence
$$\mathcal{U}G \cong \ker(r_x) \oplus \operatorname{im}(r_x).$$
The two projective summands are not trivial, therefore has nontrivial $\mathcal{U}G$-dimension in $(0,1)$. By Additivity of $\mathcal{U}G$-dimensions $\dim_{\mathcal{U}G}(\ker r_x) \in (0,1)$.

By the *non-commutative Cramer's Rule*, given $x \in \mathcal{D}G$ there exists invertible matrices P, Q over $\mathcal{U}G$ and a matrix A over $\mathbb{Z}G$, such that
$$\begin{pmatrix} x & 0 \\ 0 & I \end{pmatrix} = PAQ$$

for some identity matrix I. Therefore

$$\dim_{\mathcal{U}G}(\ker r_x) = \dim_{\mathcal{U}G}(\ker(PAV)) = \dim_{\mathcal{U}G}(\ker A) = \dim_{\mathcal{N}G}(\ker A)$$

(Implicitly using the fact that $\mathcal{U}G$ is flat over $\mathcal{N}G$ for the last equality) By assumption of the Atiyah Conjecture, this is an integer, a contradiction. This proves that $x \in \mathcal{D}G$ is invertible, hence $\mathcal{D}G$ is a field. □

Example 2.16. We have identified $\mathcal{N}\mathbb{Z}^n$ with $L^\infty(T^n)$ in Example 1.13. An operator $g \in \mathcal{N}\mathbb{Z}^n$ is injective if and only if $g \in L^\infty(T^n)$ is a.e. nonzero. The algebra of affiliated operators is

$$\begin{aligned}\mathcal{U}G &= \{fg^{-1} : T^n \to \mathbb{C} \mid f, g \in L^\infty(T^n) \text{ and } g \neq 0 \text{ a.e.}\} \\ &= \{\text{measurable functions } T^n \to \mathbb{C}\}.\end{aligned}$$

Finally,

$$\begin{aligned}\mathbb{Z}\mathbb{Z}^n &= \{\text{polynomial functions } p : T^n \to \mathbb{C}\}, \quad \text{and} \\ \mathcal{D}\mathbb{Z}^n &= \{\text{rational functions } p/q : T^n \to \mathbb{C}\}.\end{aligned}$$

Since $\mathcal{D}\mathbb{Z}^n$ is a field, we conclude that $\mathbb{Z}^n$ satisfies the Atiyah Conjecture.

2.5. **Groups extended by integers.**

Theorem 2.17. *Suppose we have an exact sequence of groups*

$$1 \to K \to G \to \mathbb{Z} \to 1$$

such that K is torsion-free and satisfies the Atiyah Conjecture. Then G is torsion-free and satisfies the Atiyah Conjecture.

This directly follows from Theorem 2.14 and the following Proposition:

Proposition 2.18. *Choose $t \in G$ which maps to the generator of $\mathbb{Z}$. Then $\mathcal{D}G$ is the field of fraction of the polynomial ring $\mathcal{D}K[t^\pm]$.*

Proof. Choose $t \in G$ which maps to the generator of $\mathbb{Z}$. It is easy to see that

$$\mathbb{Z}G = \mathbb{Z}K[t^\pm]$$

is the "polynomial ring" generated by $\mathbb{Z}K$ and t. Note that since the coefficient in $\mathbb{Z}K$ is not commutative with t, the multiplication law is as follows: for any $a_i, b_j \in \mathbb{Z}K$,

$$\begin{aligned}\sum_{i\in\mathbb{Z}} a_i t^i \cdot \sum_{j\in\mathbb{Z}} b_j t^j &= \sum_{i,j\in\mathbb{Z}} a_i t^i b_j t^j \\ &= \sum_{i,j\in\mathbb{Z}} (a_i t^i b_j t^{-i}) \cdot t^{i+j}.\end{aligned}$$

This non-commutative polynomial ring structure is sometimes called a *crossed product* of the ring $\mathbb{Z}K$ and a group $\langle t\rangle$. Similarly, the polynomial ring

$$\mathbb{Z}G \subset \mathcal{D}K[t^\pm] \subset \mathcal{U}G$$

is generated by the subring $\mathcal{D}K \subset \mathcal{U}K \subset \mathcal{U}G$ and $t \in \mathbb{Z}G \subset \mathcal{U}G$. The proof proceeds as follows:

- By assumption, $\mathcal{D}K$ is a field. Therefore $\mathcal{D}K[t^\pm]$ is a PID.

- Any PID is an Ore domain. Namely, if R is a PID, then $xR \cap yR \neq \{0\}$ for any $x, y \in R \setminus \{0\}$. (Choose $z \in R$ such that $xR + yR = zR$ and write $x = zx_1$, $y = zy_1$. There are $a, b \in R$ with $z = xa + yb$. Right multiplying by x_1 we have hence $x(1 - ax_1) = ybx_1$.)
- Therefore, the Ore localization of $\mathcal{D}K[t^{\pm}]$ exists and is a field.
- Any non-zero element of $\mathcal{D}K[t^{\pm}]$ is invertible in $\mathcal{U}G$.

  Proof. Without loss of generality, we prove that $x = a_0 + a_1 t + \cdots + a_n t^n$ is invertible in $\mathcal{U}G$, where $a_0 \in \mathcal{D}K$ is nonzero. Since $\mathcal{U}K$ is the Ore localization of $\mathcal{N}K$ over non-zero-divisors, we can write

  $$x = s^{-1} \cdot (b_0 + b_1 t + \cdots + b_n t^n), \quad b_i \in \mathcal{N}K \text{ are non-zero-divisors.}$$

  In fact $(b_0 + b_1 t + \cdots + b_n t^n)$ is a non-zero-divisor in $\mathcal{N}G$ by [Lin92, Theorem 4]. Hence x is a non-zero-divisor in $\mathcal{U}G$. □

- The localization of $\mathcal{D}K[t^{\pm}]$ embeds in $\mathcal{U}G$, and exactly equals $\mathcal{D}G$.

In conclusion, $\mathcal{D}G$ is a field and G satisfies the Atiyah Conjecture. □

Corollary 2.19. *The poly-$\mathbb{Z}$ groups satisfy the Atiyah Conjecture.*

We use the special properties of free groups and the Approximation Theorem to show that free groups satisfy the Atiyah Conjecture.

Let F be a finitely generated free group. Let

$$F = F_0 \triangleright F_1 \triangleright \cdots \triangleright F_n \triangleright \cdots$$

be the lower central series, where $F_n = [F, F_{n-1}]$. By the classical Magnus embedding of F into non-commutative power series (c.f. [LS77, I.10]), we know that:

- $\cap_n F_n = \{1\}$, and
- F_n / F_{n+1} is a free abelian group.

Therefore, it is easy to show that F/F_n is poly-$\mathbb{Z}$. Now for any matrix A over $\mathbb{Z}F$. Apply the approximation theorem [Lüc02, Theorem 13.3] to the cofinal tower of quotients:

$$F \to \cdots \to F/F_n \to \cdots \to F/F_2 \to F/F_1,$$

we know that

$$\dim_{\mathcal{N}G}(\ker A) = \lim_{n \to \infty} \dim_{\mathcal{N}(F/F_n)}(\ker A_n)$$

where A_n is the image of A in $\mathbb{Z}[F/F_n]$. Since F/F_n is poly-$\mathbb{Z}$ and satisfies the Atiyah Conjecture, $\dim_{\mathcal{N}G}(\ker A)$ is the limit of a sequence of integers, hence must be an integer. Therefore, we have proved the following:

Theorem 2.20. *Any free group satisfies the Atiyah Conjecture.*

3. Lecture 3

3.1. **Universal L^2-torsion.** In the following we will assume that G is a torsion-free group satisfying Atiyah Conjecture. We know from Linnell's theorem that $\mathcal{D}G$ is a field.

Suppose X is a cocompact, free G-CW complex with trivial L^2-Betti numbers. Equivalently, the $\mathcal{D}G$-chain complex

$$\mathcal{D}G \otimes_{\mathbb{Z}G} C_*(X)$$

is $\mathcal{D}G$-exact.

After choosing a G-basis for the cells of X, we can identify $\mathcal{D}G \otimes_{\mathbb{Z}G} C_*(X)$ with a free $\mathcal{D}G$-module. This exact $\mathcal{D}G$-chain complex can be decomposed as

$$\begin{tikzcd}
\mathcal{D}G^{r_{n+1}} \arrow[dr, "A_{n+1}", "\cong"'] & \mathcal{D}G^{r_n} \arrow[dr, "A_n", "\cong"'] & \mathcal{D}G^{r_{n-1}} \\
\oplus & \oplus & \oplus \\
\mathcal{D}G^{r_{n+2}} & \mathcal{D}G^{r_{n+1}} & \mathcal{D}G^{r_n} \\
\| & \| & \| \\
\mathcal{D}G \otimes_{\mathbb{Z}G} C_{n+1}(\widehat{M}) \arrow[r, "\partial_{n+1}"] & \mathcal{D}G \otimes_{\mathbb{Z}G} C_n(\widehat{M}) \arrow[r, "\partial_n"] & \mathcal{D}G \otimes_{\mathbb{Z}G} C_{n-1}(\widehat{M})
\end{tikzcd}$$

Definition 3.1 (Universal L^2-torsion)**.** The *universal L^2-torsion* of X is defined as

$$\tau_u^{(2)}(X) := \prod_{n\in\mathbb{Z}} \det(A_n)^{(-1)^n}.$$

Here $\det(A)$ means the element of $K_1(\mathcal{D}G)$ represented by the invertible matrix A. Alternatively, it can be understood as the *Dieudonné determinant*

$$\det : \mathrm{GL}(\mathcal{D}G) \to K_1(\mathcal{D}G) \cong (\mathcal{D}G)^\times_{\mathrm{ab}}.$$

Finally, the universal L^2-torsion $\tau_u^{(2)}(X)$ is well-defined (i.e. independent of the cellular basis of X) in the *Whitehead group*

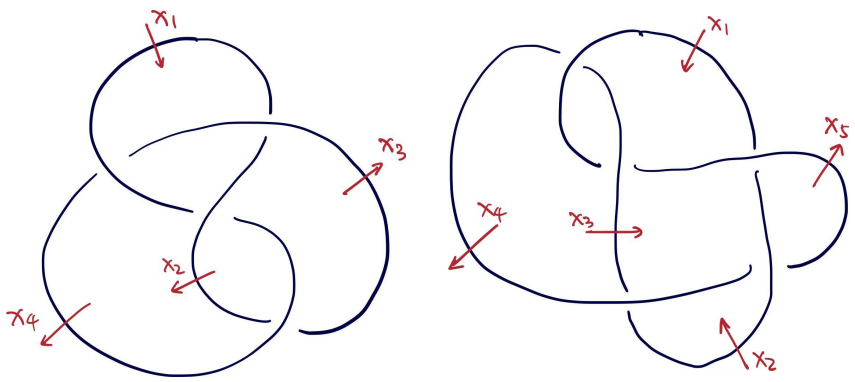
$$\mathrm{Wh}(\mathcal{D}G) := K_1(\mathcal{D}G)/\pm G.$$

Example 3.2. For M_1 the 4_1-knot complement, we have

$$\Delta(M_1) = (1 - 3t + t^2),$$
$$\tau_u^{(2)}(M_1) = [1 - (x_2 + x_3 + x_4) + x_4x_3] \cdot [1 - x_4]^{-1}.$$

For M_2 the 5_2-knot complement, we have

$$\Delta(M_2) = (2 - 3t + 2t^2),$$
$$\tau_u^{(2)}(M_2) = \left[(1 + x_4^{-1}x_3) - (x_1 + x_5 + x_2) + (x_5x_1 + x_2x_1)\right] \cdot [1 - x_5]^{-1}.$$



Definition 3.3 (Leading term map)**.** Suppose $\phi : G \to \mathbb{Z}$ is a nontrivial homomorphism whose kernel is K. Choose $t \in G$ such that $\phi(t)$ generates the image of ϕ. Identify

$$\mathcal{D}G = \mathcal{D}K[t^\pm] \cdot \mathcal{D}K[t^\pm]^{-1}.$$

For any $x \in \mathcal{D}G$, write

$$x = p \cdot q^{-1}, \quad p, q \in \mathcal{D}K[t^\pm],$$

define $L_\phi : \mathcal{D}G \to \mathcal{D}G$,

$$L_\phi(x) := (\text{lowest } t\text{-degree term of } p) \cdot (\text{lowest } t\text{-degree term of } q)^{-1}.$$

This induces homomorphisms

$$L_\phi : K_1(\mathcal{D}G) \to K_1(\mathcal{D}G) \quad \text{and} \quad L_\phi : \mathrm{Wh}(\mathcal{D}G) \to \mathrm{Wh}(\mathcal{D}G).$$

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